Lesson for the course "Human-Computer Interaction"

Introduction to the discipline of Human-Computer Interaction

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Who am I?

- I am a researcher in HCI
  - Music Technology
  - Multisensory interactive systems



Smart Musical Instruments

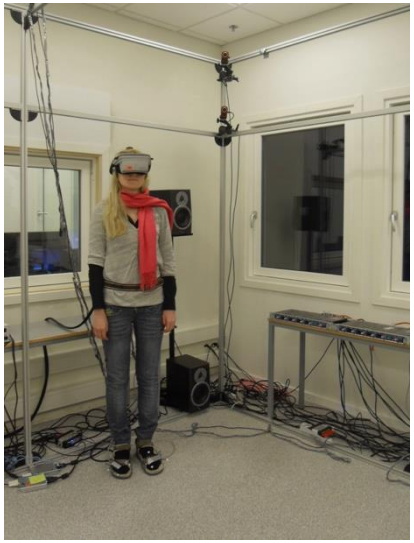


The Internet of Musical Things









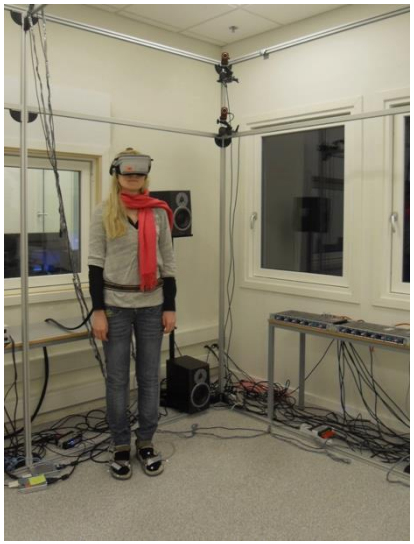
Audio-haptic shoes for virtual reality or rehabilitation



Videos of the systems in action



https://www.youtube.com/watch?v=fqzEQnsSIoY



https://www.youtube.com/watch?v=kRKcKgYCPCY

Who am I?

- I create novel systems and I study the interaction the users have with them
- For instance, this means
  - To design the systems **involving the users**
  - And evaluate the systems in their **context** of use





My research interests

- HCI for multisensory interactive systems
- Sound and music computing
- Haptic technology
- Virtual reality
- Internet of Things
- Semantic Web
- Affective computing
- Embedded systems

- **I have several thesis proposals, especially for developers**

What you will study in this course

- The course has the general objective of providing you with the **fundamental notions** about the interaction between humans and computerized systems
- In particular, the course is focused on the basic principles of **usability** and on their application during the **design** and **evaluation** of interfaces of various types and for various domains
- There will be 24 lessons
  - Frontal lessons
  - Laboratories
  - Seminars held by **external experts**

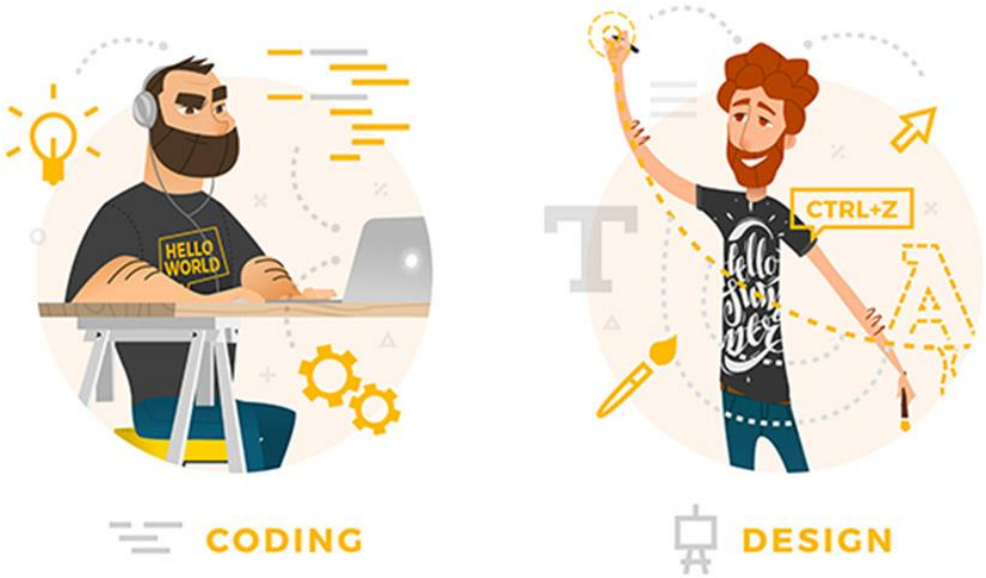
Online material

- My lessons are also held online in a synchronous mode
- Zoom link: https://unitn.zoom.us/j/81967761768
  Meeting ID: 819 6776 1768 Passcode: 432707
- The lessons are also recorded, and made available on the Google Drive folder:
  https://drive.google.com/drive/folders/15_3IpH9uMph5SF8koqFHNoBYVePU2WXX?usp=sharing

NOTE: DO NOT USE YOUR OWN GMAIL ACCOUNT! The link is shared with your unitn account. I will not release access to requests coming from gmail accounts

What you will study in this course

- The focus is on
  - The study of the interaction
  - Design
  - Evaluation
- The focus is not on the implementation
  - **No programming is involved**



Upon successful completion of this module, you should be able to:

- Design and **critically** assess the user interface of an interactive system as well as the interaction modalities afforded by it
- Use **tools**, **techniques** and **models** for the design and evaluation of the interaction

  axure https://www.axure.com

  - Axure RP is arguably the best **interaction design tool** on the market (free for students!): https://www.axure.com/edu
  - Download it and register an account as a student
- **Understand the needs and the role of the user** and relate them to the design of usable interactive systems

Upon successful completion of this module, you should be able to:

- Know the fundamentals of the human **cognitive processes** and their implications for the design of interactive interfaces
- Develop **prototypes** of interfaces of websites and mobile apps, considering usability aspects as well as the peculiarities of the interaction
- You will do a group **project** and learn how to work in team
- Understand the issues of individuals with physical and/or technological disabilities related to the **accessibility** of interactive systems

Syllabus

- Lesson 1: **Introduction to the discipline of HCI**
- Lesson 2: **The process of Interaction Design & the User Experience**
- Lesson 3: **PACT analysis & Benchmarking**
- Lesson 4: **Sensors, Actuators and Types of Interactions**
- Lesson 5: **HCI, AI, and IoT**
- Lesson 6: **Data gathering and analysis**
- Lesson 7: **Data interpretation and presentation + Feedback about projects**

Syllabus

- Lesson 8: **Discovering requirements**
- Lesson 9: **Workshop on data gathering and analysis techniques**
- Lesson 10: **Design and prototyping + introduction to Axure**
- Lesson 11: **Design principles and heuristics**
- Lesson 12: **Axure**
- Lesson 13: **Workshop on tools for sketching + tips to write the report for the exam**

Syllabus

- Lesson 14: **Evaluation methods**
- Lesson 15: **Visual Perception part 1**
- Lesson 16: **Axure RP**
- Lesson 17: **Visual Perception part 2, Reading**
- Lesson 18: **Attention and memory**
- Lesson 19: **Learning**
- Lesson 20: **Time, Space, and eye-hand coordination**
  - You will see a couple of mathematical equations

Syllabus

- Lesson 21: **Feedback about projects**
- Lesson 22: **Emotional design**
- Lesson 23: **Infographics** by Maurizio Napolitano from FBK
- Lesson 24: **Feedback about projects** and tips about how to write the report

- **I strongly recommend to work on the project during the course, so you can receive feedback on your project before the exam**
- The feedback I will give is directed to the whole class, not only to you, so everybody can learn from your doubts

Questions and answers

- There will be in total **3 lessons where you will receive feedback about your project**
  - Only if you want it
- Those lessons will consist of answering to any sort of questions you may have about the project: the feedback about a project will be given **to all the class**
- You can always ask (publically) for a feedback **at the end of the lesson**, the answer is given to all class
- Always, all feedback I will give is directed to the whole class, not only to you (so everybody can learn from your question)

Questions and answers

- The slides should contain the answer to all your doubts
- There is always the reference teaching material that you can consult (books, scientific articles, weblinks)
- **Online FAQ document continuously updated** (it will be updated even after the end of the course)
  - https://docs.google.com/document/d/1mxbOxMN8RCfryTwVSmlUt3Nv5kHk4tUVTQ7fpbhC0fE/edit?usp=sharing
  - If you write me an e-mail with a question, **you need to formulate the question** (in English) **with a general character so that your question can be of interest for all the students** (not something specific to your own project)
  - Then I will report the answer in the FAQ document (I will update it weekly with the answers to the questions)
  - Please: consult the FAQ document before contacting me

Teaching material

- The slides I provide you contain a summary of what you need to know to gain knowledge about the HCI field and pass the exam successfully
  - They are based on books, scientific articles, and web pages
- **The slides of a lesson will be uploaded before the lesson begins**
- Readings of the links, articles, and book chapters indicated in the ***Bibliography slides*** is recommended

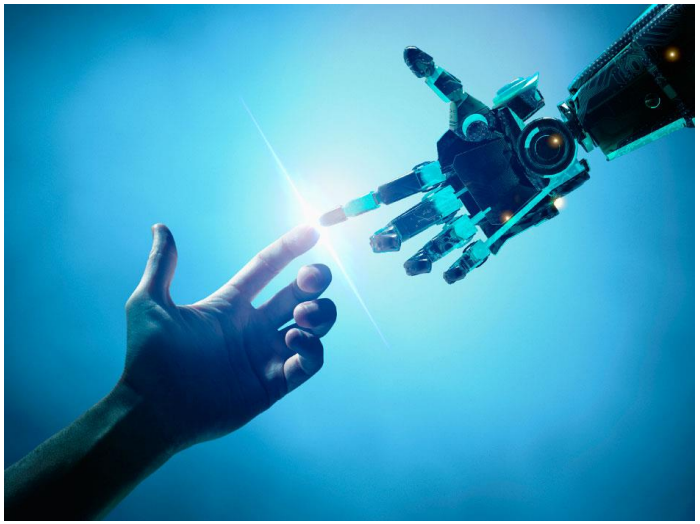
Teaching material

- Books
  - Sharp, H., Rogers, Y. & Preece, J. (2019). *"Interaction Design: beyond human-computer interaction"*. John Wiley & Sons, Inc. – 5th Edition, 2019. http://www.id-book.com/
  - Jeff Johnson (2014). *"Designing with the Mind in Mind: Simple Guide to Understanding User Interface Design Guidelines"* - 2nd Edition
  - Books are available at the library in both paper and digital forms
- Websites links and scientific articles indicated in the slides
  - https://www.interaction-design.org/
  - https://uxdesign.cc/
  - https://www.nngroup.com
  - The ACM Conference on Human Factors in Computing Systems (CHI)

The discipline of HCI

- Fundamentally, HCI is the study of interaction between people and computers with the aim to understand and improve those interactions

- *"Human-computer interaction is a discipline concerned with the design, evaluation and implementation of interactive computing systems for human use and with the study of major phenomena surrounding them."*
  – ACM SIGCHI, 1992



Disciplines contributing to HCI

- The field of HCI covers a wide range of topics, and its development has relied on contributions from many disciplines

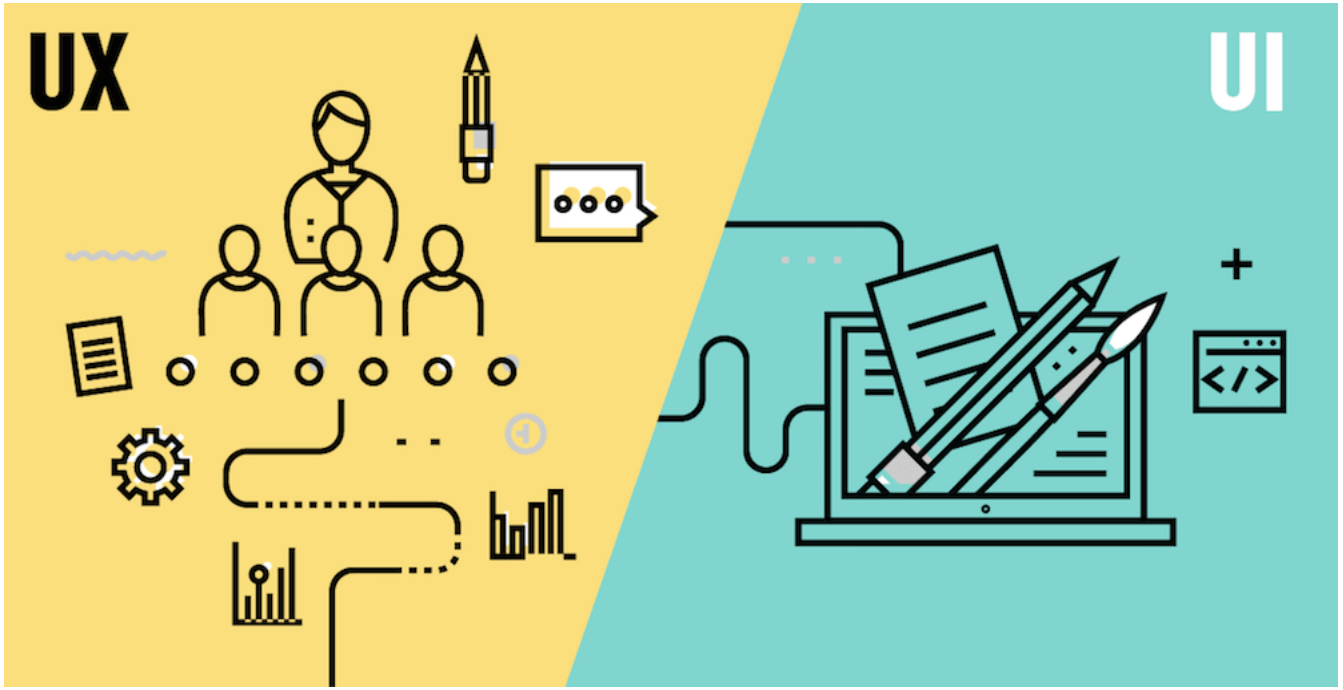
HCI {
- Computer science
- Design
- Engineering
- Psychology
- Sociology
- Philosophy
- Human factors (Ergonomics)
- Anthropology

- …a **very interdisciplinary** field

Interaction design (IxD)/UX design

- *"Creating interactive products to support people in their everyday and working life"* (from the book "Interaction Design" – 5th edition)
  - Basically, creating experiences which fit, extend, and enhance the way people work, communicate and interact
- The term "**User Experience Design**" is used more widely in **industry** to refer to the profession
  - However, the two terms can be used interchangeably
- "**Human Computer Interaction**" is the term to indicate the **academic discipline**

UX design vs UI design



- UI ("User Interface") design is often confused with UX design
  - UI design is more concerned with the surface and overall feel of a design
  - UX covers the *entire spectrum* of the user experience

UX design and UI design

- UI design typically refers to graphical user interfaces (GUIs) but also includes others, such as
  - Voice-controlled interfaces: involve oral-auditory interaction
  - Gesture-based interfaces: witness users engaging with 3D design spaces via bodily motions





- In GUIs, the UI consists of the buttons users click on, the text they read, the images, sliders, text entry fields, and all the rest of the items the user interacts with (e.g., screen layout, transitions, interface animations)

UX design and UI design

- UX designers are also concerned with an application's user interface, and this is why people get confused about the difference between the two
- But whereas UI designers are tasked with deciding how the user interface will **look**, UX designers are in charge of determining how the user interface **operates**
- **UX designers determine the structure of the interface and the functionality. How it's organized and how all the parts relate to one another**
  - In short, they design how the UI works

What constitutes good design?

- Good design is one that fills the gap between business goals and user needs
- To fill this gap, a process must be followed: a process that takes into consideration best practices of user experience and usability guidelines to produce the desired outcome
- **A good designer knows how to get into the mindset of his users, and turns their needs into a meaningful, desirable, and easy-to-use product or service**
- *"Good design is actually a lot harder to notice than poor design, in part because good designs fit our needs so well that the design is invisible."- Don Norman*

Good design

- Why is the TiVo remote much better designed than standard remote controls?
  - Peanut shaped to fit in hand
  - Logical layout and color-coded, distinctive buttons
  - Easy to locate buttons



Bad design



- Stove top control design
- Example of a thing that is hard to use because it does not follow human factors principles
- The problem is that it is difficult to tell which control goes with which burner



- Good design: the solution is to arrange the controls in the same configuration as the burners
- It is quite easy to tell which burner goes with which control

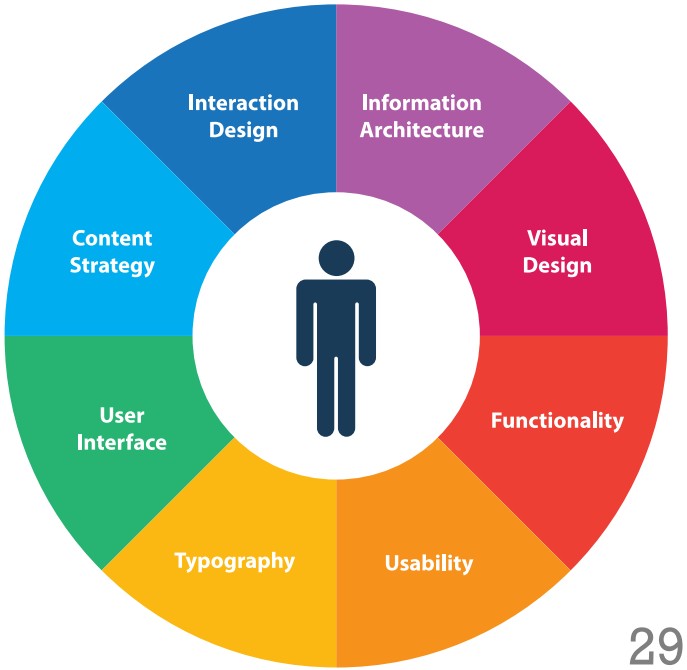
Bad design

- Alan Cooper (2018), a well-known user experience (UX) guru, bemoans the fact that much of today's software suffers from the same interaction errors that were around 20 years ago
  - Many interfaces of new products do not adhere to the interaction design principles validated in the 1990s
  - E.g., he notes that many apps do not follow even the most basic of UX principles, such as offering an "undo" option
  - He exclaims that it is *"inexplicable and unforgivable that these violations continue to resurface in new products today"*

- In this course you will see many examples of good and bad designs for user interfaces

How to achieve a good user experience?

- We need to be able to understand how to **reduce the negative aspects** of the user experience (e.g., frustration and annoyance) while **enhancing the positive ones** (e.g., enjoyment and efficacy)
- This entails developing interactive products that are easy, effective, and pleasurable to use from the users' perspective
- Paradigm shift:
  - Technology-driven design
    - Technology influences the design of the interface
  - **User-centered design**
    - Users' requirements define the functionality of the interface



The goals of HCI

- **To improve the interaction between users and computers** by making computers more user-friendly and receptive to the user's needs
- To develop or improve the safety, utility, effectiveness, efficiency, and usability of systems that include computers
- To design systems that **minimize the barrier** between the human's cognitive model of what they want to accomplish and the computer's understanding of the user's task



Improving the usability is a key factor

- Underlying the whole theme of HCI is the belief that **people** using a computer system should come **first**
- Their **needs, capabilities and preferences** for conducting various tasks should direct the design of the interactive systems
  - People should not have to change the way that they use a system in order to fit in with it
  - Instead, the system should be designed to match their requirements



Importance of HCI

- Today, the User Interface (UI) is often one of the **most critical factors** regarding the success or failure of a computer system
- Badly designed interfaces can lead to serious implications
- **Bad design costs money, time, and lives**
- If you build poor interfaces you might find
  - The company using it loses money since its workforce is **less productive** than it could be
  - The **quality of life of the users** who use your system is reduced (low consumer satisfaction)
  - Disastrous and possibly **fatal errors** happen in systems that are safety-critical

Importance of HCI

- Technological systems are an integral part of our daily lives, so **it is important to understand and learn how the design of a particular system can affect perception and behaviour**
- Smart devices (e.g., mobile phones, PDAs), smart products (e.g., car) and smart environments (e.g., ambient intelligence, smart buildings) **are enabling new services that require innovative interfaces**
- HCI focuses on the study, design, development and evaluation of novel user interfaces, interactive systems and services, with the goal of improving the users' interaction with them

Importance of HCI

- There is a widespread and ubiquitous use of computerized systems
  - This widespread use makes **design choices** very important: millions of people might suffer because of bad design
- A user who does not have a good experience with an application is a potential customer that your company is at risk of losing





Interaction design in business

- Large number of ID consultancies, examples of well known ones include:
  - **Nielsen Norman Group**: *"help companies enter the age of the consumer, designing human-centered products and services"*
  - **Cooper:** *"From research and product to goal-related design"*
  - **IDEO:** *"creates products, services and environments for companies pioneering new ways to provide value to their customers"*

HCI professional roles today

- As HCI is so important, there is a **growing need of HCI professionals**
  - Huge request for this kind of expertise
  - The average UX designer salary is increasing

- Various job titles…
  - UX designer
  - Interaction designer
  - User Interface designer
  - User Interaction designer
  - HCI designer
  - Usability engineer
  - Usability consultant
  - Usability analyst
  - Information architect
  - User interface researcher



What does a UX designer design?

- For example:
  - A new version of an existing interactive technology
    - E.g., re-design of ticket machines for public transport
  - An interactive technology for an existing human activity
    - E.g., the Kindle for the activity of "reading books"
  - A new human activity thanks to a new technology
    - E.g., Instagram stories

The UX designer role

- It is not all about designing a UI (e.g., of an app)
  - Yes, you have to deliver some prototypes but…
- UX design is a **highly collaborative role**
  - You need to meet with company executives to understand their vision
  - You may rely on the customer service team for valuable user insights
  - Importantly, the role entails to do user research: you have to be in touch with users





The UX designer role

- Designers need to **know** many different things about users, technologies, and the interactions among them to create effective user experiences
- At the least, they need to understand how people act and react to events and how they communicate and interact with each other
- To be able to create engaging user experiences, they also need to understand how emotions work, what is meant by aesthetics and desirability
- They also need to understand the business side, technical side, manufacturing side, and marketing side

Common challenges UX designer face… and how to overcome them

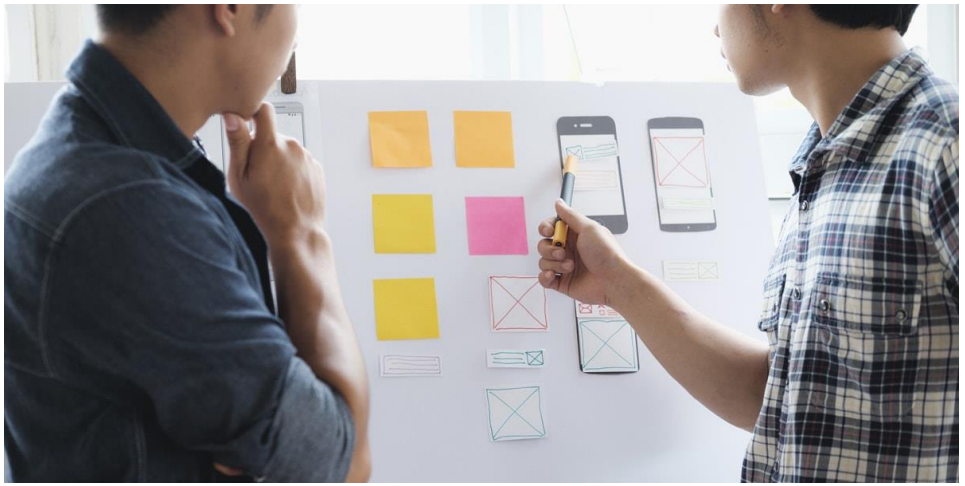
- It's an exciting time to be a UX designer
- …but there are also some challenge
  - **Misconceptions about what UX designers do**
  - **Conducting research under time and budget constraints**
  - **Bridging the gap between design and development**



Misconceptions about what UX designers do

- One of the most common challenges you'll face as a UX designer is a **lack of understanding** about what you do
- The multidisciplinary nature of UX is often underestimated
  - For many people, the term user experience design is synonymous with just "making things look good"
  - Some stakeholders may have no idea of what you're here for. They may think they already know what their users want. They just want the thing they've made looks better
- At the same time, you might find that the business value of UX is overlooked

Misconceptions about what UX designers do



- An example
  - Imagine a scenario where you're brought on board to design an app
  - The company already knows who their users are and what they want; all you need to do is take care of the design
  - **However, during the research phase, you uncover new insights about the user**
  - You now need to take the project in **a completely different direction** — and you need to convince the CEO and other stakeholders that it's absolutely necessary
  - If they were simply expecting you to create some prototypes, they may not respond well to this revelation…

Misconceptions about what UX designers do: solutions

- As a UX designer, it's important to **set expectations very early on**
  - Start each new project with a stakeholder kick-off, explaining what UX is, what it is not, what kind of activities you'll be doing and why
  - As you move through the project, provide regular updates and opportunities for your colleagues to ask questions
  - A well-prepared stakeholders kick-off and some one-to-one interviews should clear up the confusion: UX Design is far more than just "making software look good"

    - **You will see this during this module, and while doing the group project**

Conducting research under time and budget constraints





- Situation where there is no time and money for conducting user research
  - Your client or manager might even ask you to skip the research phase altogether…

- It's a common paradox
  - The decision-makers are convinced they know all there is to know about their customers
  - With limited resources, it would make sense to start designing right away
  - **As a UX designer, however, you know the opposite to be true:** user research is crucial for validating assumptions and making sure you're solving a real user problem
  - **Skipping user research has been known to cost companies millions**

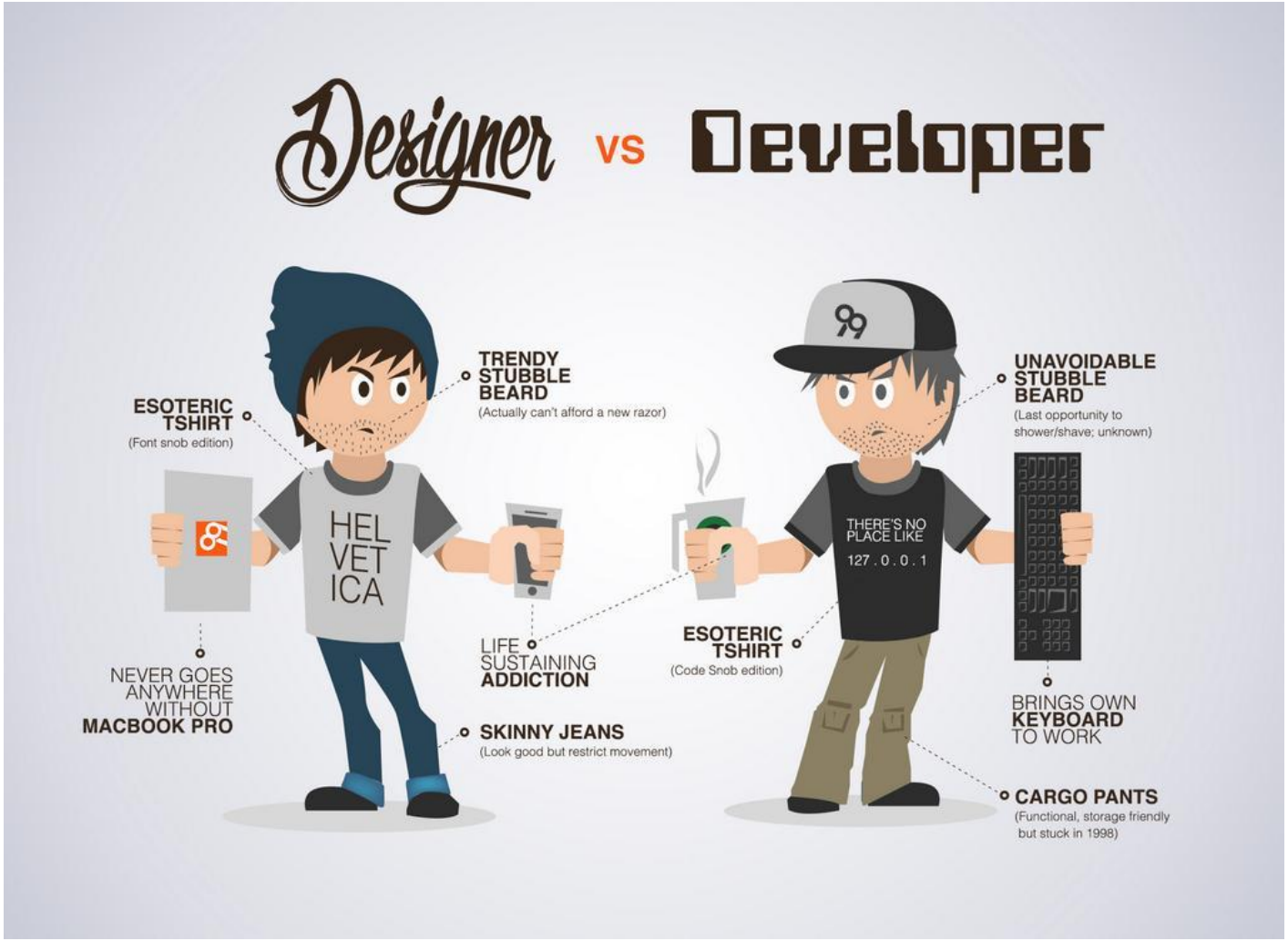
Conducting research under time and budget constraints: solutions

- Bypassing the initial user research phase **is not an option**
- This is a path that many UX designers have walked upon before, and there are plenty of tried-and-tested methods for conducting quick yet effective research on a budget
  - Focus on pooling the resources you've got and optimizing the process as much as possible
  - If you need to recruit participants for user interviews, for example, put the word out on social media and conduct interviews over the phone rather than in person
- When faced with time and budget constraints, it is important to remember that **some research is better than none** – even if it's not as thorough as you would like

Designers and developers

- Designers and developers have very different yet equally important roles to play in the product development process
  - **Without Development, Design is useless**
  - **Without Design, Development is unusable**
- Collaboration between the two is paramount, but
  - Designers and developers don't always get along too well
  - They can have two different mindsets, backgrounds, and approaches to work, which commonly leads to not seeing eye-to-eye!

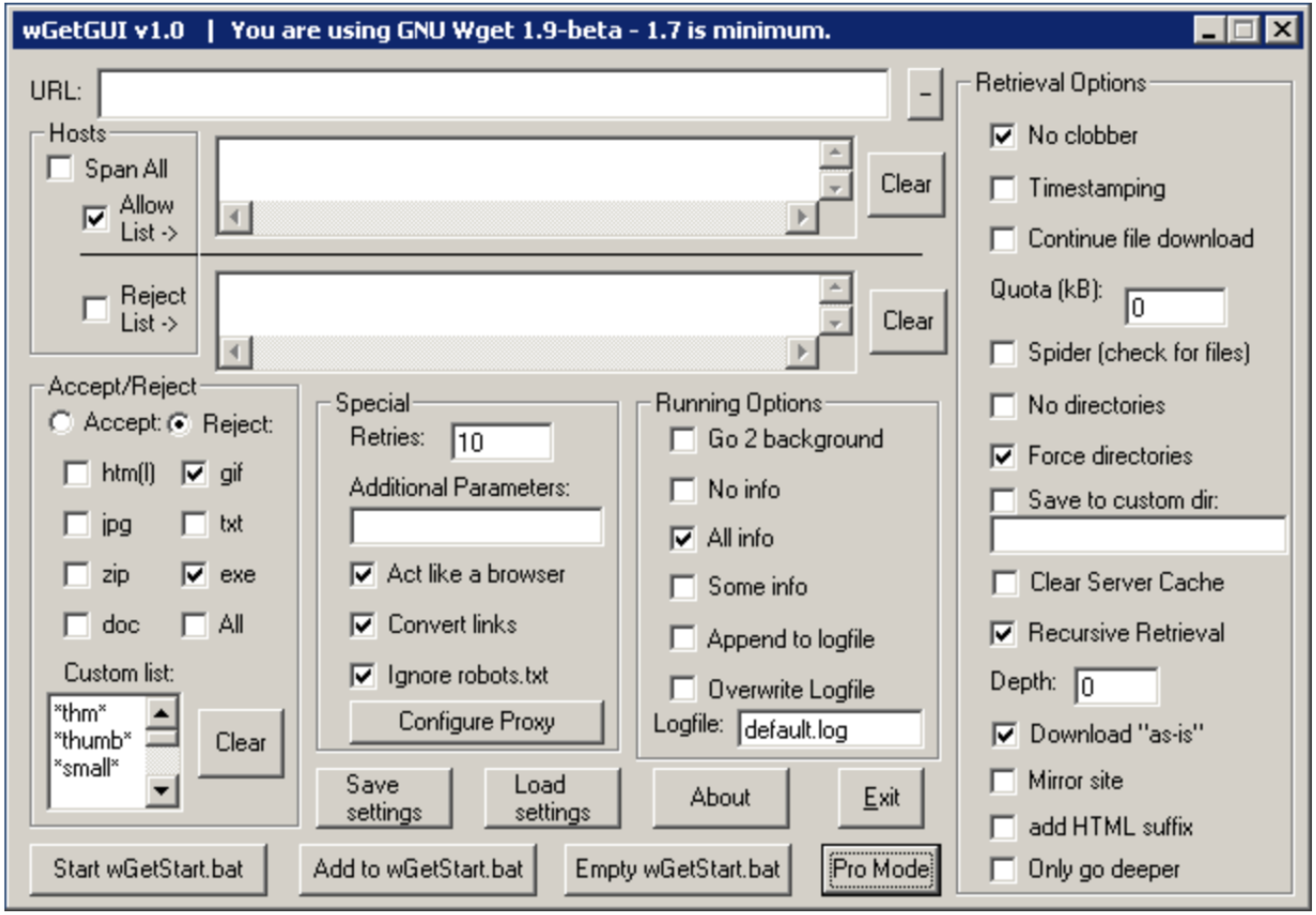




- When designers come up with a new way to improve a website's UX, developers explain how it increases page loading speed…
- And sometimes, developers come up with their own design concepts that designers struggle to accept…

This Is What Happens When You Let Developers Create UI

Deep down inside every software developer, **there's a budding graphic designer waiting to get out.** And if you let that happen, you're in trouble. Or at least your users will be, anyway:



The approach of UX designers to solving problems is different to that of developers

- **Design is user-oriented**
  - designers' focus is on the people and how every aspect of the product will affect them
- **Development is solution-oriented**
  - developers' focus is on the problem's constraints and how to eliminate what is not going to be a part of the solution
- Design's aim is to accommodate a **large number of factors** within a single solution (visual appeal, accessibility, usability, marketing, branding, historical and geographical context, etc.) so that the largest possible number of customers will understand and use the solution
- Development's aim is to direct the complexity and subtlety of a given **algorithm** towards the same large number of customers so that they will use it as **efficiently and effectively** as possible

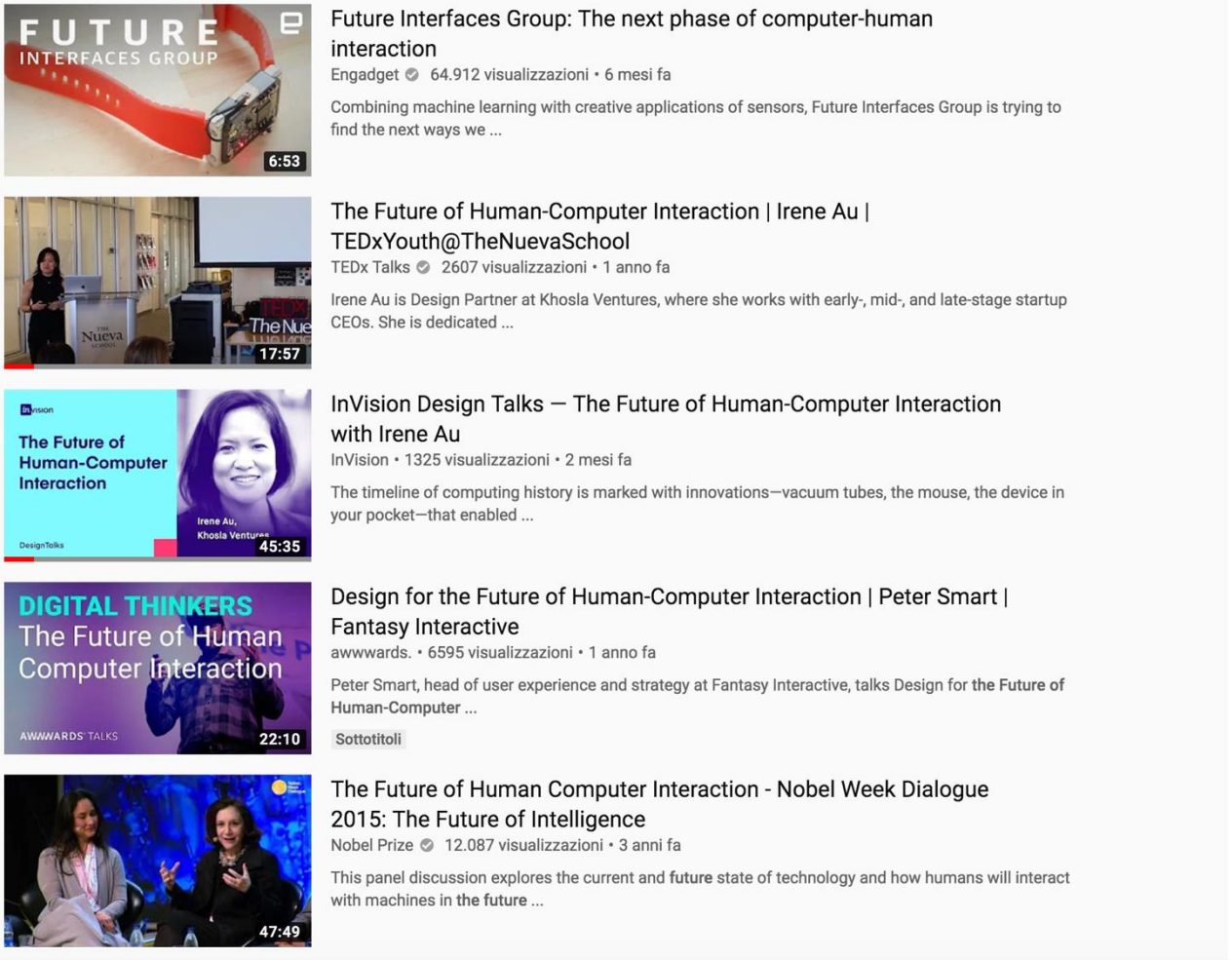
The approach of UX designers to solving problems is different to that of developers

- The funny part is, **both designers and developers will argue that theirs is the way to solve a problem!**
  - Developers will say that focusing on the people is not literally solving the problem
  - Designers will say that the problem is not really the problem
  - Both are right and both need to be right
- When designers and developers aren't aligned, both the process and the product suffer
  - Lack of communication and knowledge gaps tend to be the main culprits, resulting in excessive (and redundant) feedback loops, different interpretations of the same objectives, and unexpected quality assurance issues

Bridging the gap between design and development: solutions

- The best way to avoid discord between design and development is to **communicate early and often**
- Involve the developers as soon as possible; this way, they'll be able to identify any potential issues long before the deadlock
- Knowledge sharing is also key
  - For UX designers it's worth taking the time to understand how the developers work and the challenges they face
  - and vice versa

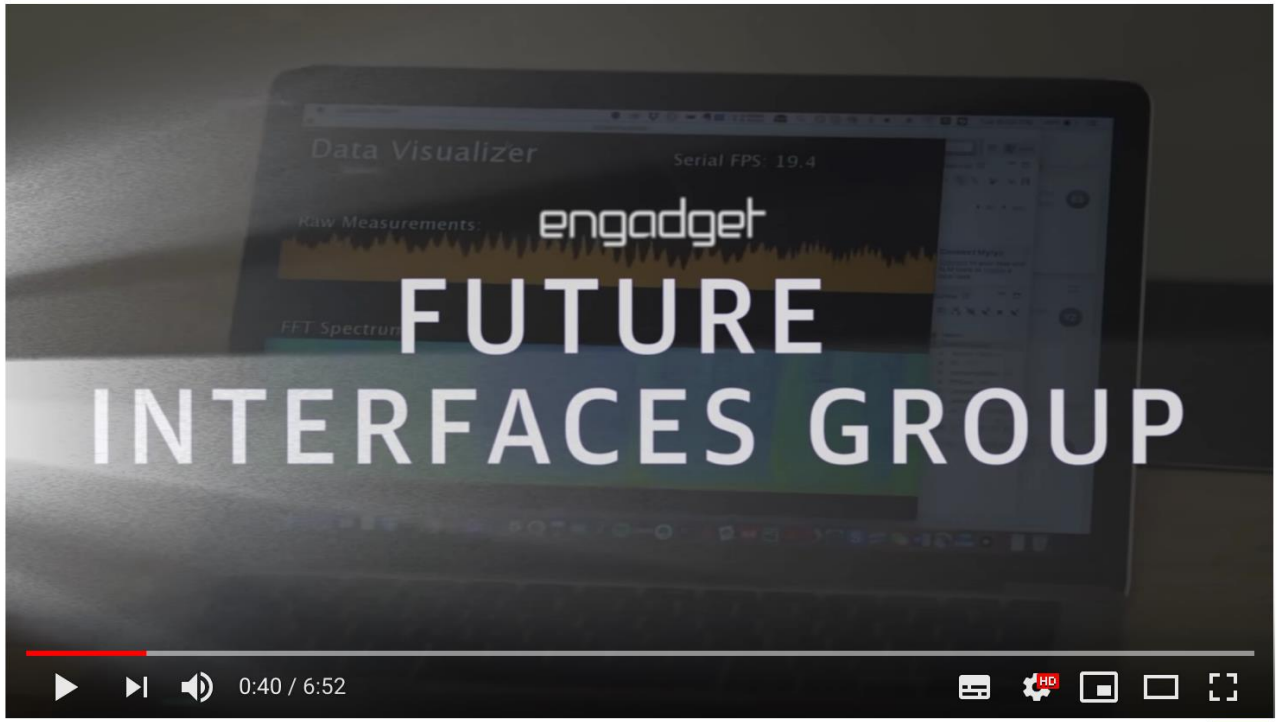
The future of HCI?



- **YOU are the future of HCI**

The future of HCI?

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Future Interfaces Group: The next phase of computer-human interaction

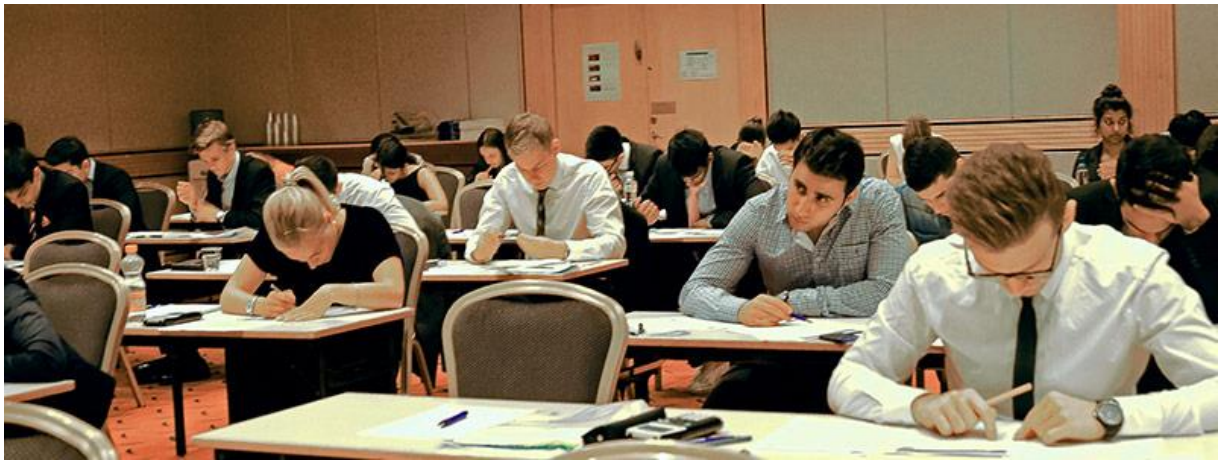
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Engadget
Pubblicato il 17 dic 2018

ISCRIVITI 852.169

Combining machine learning with creative applications of sensors, Future Interfaces Group is trying to find the next ways we'll interface with computers beyond our current modes of voice and touch. Key technologies whose interfaces are yet to be standardized include smartwatches, AR/VR, and the internet of things.

Exam



- The exam consists of **a group report on the project** (there is no a presentation or an oral exam)
- The project consists on the interaction design process of an interactive system

Exam

- You will have to work on a project **in groups of 3**
  - No groups with more than 3 members allowed
  - **Groups of 3 are the preferred way**
  - Groups of 2 members or individual project are possible **but not recommended**: interaction design is a process that is typically done in groups with several exchanges
- I strongly recommend you to start very soon in forming the groups (start already today!)
- Please make a whatsapp/telegram group where you can find members to form a group
- If during the project development a member does not work or behave well, you can always break the group and make the project individually or in 2

The project: 3 categories

Increasing level of complexity



- You can choose either:
  1. **A web site displayed on a desktop computer**
  2. **An app for a mobile device (e.g., smartphone, tablet)**
  3. **System composed by a mobile app or program in conjunction with a smart object or smart environment [*preferred*]**
- You don't have to program code
- The smart object/environment and its interactions can be simulated
- The higher the complexity of the project the higher the chance to have a high grade
- The category 3 – System, is the more complex and is the one I prefer. However, even with category 1 and 2 you can get 30L, what is important is that you demonstrate to have learned the content of the course and to be able to apply it

Examples of project topics

- **You are free to choose as well as to propose new topics**

- App, website, or system to raise awareness on:
  - Any social issue
  - Climate change
  - Food waste (e.g., in families or organizations)
  - Recycling attitudes
  - Sustainability aspects of an activity/work
  - Gender equality
  - Environment/air pollution
  - Inactive and sedentary lifestyle: how could technology help increase people's adherence to physical activity?
  - Smoking/Alcohol consumption (e.g., a stop smoking app)

Examples of project topics

- App, website, or system to facilitate/improve:
  - Mobility in Trento city or Trentino-Alto-Adige region
  - Tourist guidance across the Trento city or museums (e.g., Muse)
  - Art consumption (e.g., network of galleries/museums)
  - Sports meetings among friends (e.g., find a venue, manage times)
  - Classical music concerts participation among teenagers/ young people/ adults/ elderly population
  - Washing machines use
  - Search of eatable mushrooms in woods, targeted to mushrooms hunters (e.g., using machine learning to tell good from bad mushrooms)

Examples of project topics

- App, website, or system to monitor:
  - Physical rehabilitation progresses
  - Air/noise/water pollution (smart city)
  - Car parking availability (e.g., smart city with sensors on streets/asphalt…)
  - Electricity consumption in houses, companies, cities

Suggestion: BE ORIGINAL in the choice of your project topic, use your creativity

Exam: group report on the project

- The group **report** consists of the description of the activities performed during the project, in relation to the theoretical notions presented during the module
  - The report must be maximum **12 pages long** (excluding the index and references), **fontsize at least 11**, margins 2.5 cm
  - The report can be written in Italian or in English
- You need to include the **prototype** of the interactive system developed and a **video** of it in action
  - The duration of the video of the prototype must be **between 3 and 5 minutes**

Exam: group report on the project

- Note: **all documentation** related to the project phases must be stored in an online repository that has to be accessible to the examiner
  - Such a documentation should include audio/visual material of interviews and evaluations of prototypes, files containing the gathered data and their analysis, and the various versions of the prototypes

Criteria for the assessment of the group project and report

- Coherence and quality of the application of the theoretical concepts presented during the module
- Complexity and quality of the interactive prototype and quality of the resulting interaction in relation to the target user group for which it has been designed
- Originality, creativity, and innovation of the proposed solution
- Conceptual quality of the content, including the level of the critical reflection in relation to the topics presented during the lessons, the clarity of the motivations explanation, and the relation to the theoretical aspects presented during the lessons
- Clarity of the text, in terms of content, layout (including figures, tables) and grammar

Recommendations

- **Start early** with your project!
  - **Form the group as soon as possible**
  - **By doing the project during the course you can receive feedback about your project before the exam**



Homework

- https://www.axure.com
  - **Download it and register an account as a student**

- Select a websites and an app, and analyze them in terms of your user experience
  - Write down your analysis (bullet points)
  - Keep it: at the end of the course you will redo this exercise and you will hopefully notice the differences with your first analysis

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